SIMATS ENGINEERING

ASSESSMENT TOOL 1: Agent Design Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS

COURSE CODE: MLA01

COURSE OUTCOME COVERED

CO1: *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems.* (BTL 3)

Assessment Tool Weightage: 55 %

Total Marks: 50 **Time: 60 Mins**
No. of Questions: 1

Annexure A

Agent Design Assessment

Code of Conduct:

I D. Sri Anjaneya / 192425219 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Question	Problem Understanding & AI Concept Identification	Dialogflow Agent Design & Implementation	Problem Formulation & Conversation Flow Design	Application of AI Techniques & Reasoning	Testing, Evaluation & Documentation	Total
Weightage	20%	30%	20%	20%	10%	100%
Q1						

Signature of the Faculty Total Marks Awarded

Scenario

A university wants to develop an **AI-based Student Support Assistant** to provide instant responses to student queries related to course registration, timetable, examination schedules, attendance details, and campus services.

As an AI developer, design and implement a conversational agent using **Dialogflow** that can understand user requests, process intents, and provide appropriate responses.

Q. No.	Criteria to be evaluated	BL	Marks
1	**Problem Analysis and Agent Design:** Analyze the student support problem and explain how Artificial Intelligence can solve the problem. Design the conversational agent by defining its objective, users, environment, inputs, outputs, and expected performance measures.	BL3	10
2	**Dialogflow Agent Development:** Create a Dialogflow agent by designing appropriate Intents, Entities, Training Phrases, Responses, and Contexts. Explain how the agent perceives user queries and generates responses. Provide screenshots of the implementation.	BL5	10
3	**Problem Formulation and Decision Flow:** Represent the chatbot interaction as a problem-solving process by defining the initial state, user goal, possible actions, state transitions, and goal completion conditions. Draw the conversation flow diagram.	BL6	10
4	**Search Strategy Application:** Analyze how search techniques used in AI problem solving can support conversational agents. Apply a suitable search approach (such as Breadth First Search or heuristic-based search) to model intent selection or dialogue navigation and justify your choice.	BL5	10
5	**Testing and Evaluation:** Test the Dialogflow agent with different user queries. Evaluate accuracy, response quality, handling of unknown queries, and suggest improvements.	BL6	10

Evaluation Rubrics

Criteria	Weightage (%)	Excellent (Full Marks)	Good	Average	Poor
Problem Understanding & AI Concept Identification	20%	10 – Clearly analyzes the student support problem, defines AI application, agent objective, users, environment, inputs, outputs, and performance measures with proper justification.	7.5 – Identifies most components of the agent design with minor omissions.	5 – Provides basic problem analysis with limited explanation of agent components.	0–2.5 – Incomplete understanding of problem and agent design.
Dialogflow Agent Design & Implementation	30%	10 – Successfully creates a functional Dialogflow agent with appropriate intents, entities, training phrases, responses, contexts, and explains interaction flow with screenshots.	7.5 – Develops major Dialogflow components with minor configuration errors.	5 – Creates basic intents and responses with limited entity/context utilization.	0–2.5 – Incomplete or incorrect Dialogflow implementation.
Problem Formulation & Conversation Flow Design	20%	10 – Clearly represents chatbot interaction using initial state, goal state, actions, state transitions, goal conditions, and a well-designed conversation flow diagram.	7.5 – Provides a structured flow with minor missing details.	5 – Provides basic flow representation with limited explanation.	0–2.5 – Poor or missing problem formulation and decision flow.
Application of AI Techniques & Reasoning	20%	10 – Correctly analyzes and applies suitable search strategies (BFS/heuristic search) for intent selection or dialogue navigation with strong justification.	7.5 – Applies appropriate search strategy with reasonable explanation.	5 – Provides basic understanding of search strategy application.	0–2.5 – Unable to relate search concepts with conversational agent.
Testing, Evaluation & Documentation	10%	10 – Performs comprehensive testing using multiple user queries, evaluates accuracy, response quality, unknown query handling, and suggests meaningful improvements.	7.5 – Performs adequate testing and provides evaluation with minor gaps.	5 – Conducts limited testing with basic observations.	0–2.5 – Insufficient testing and evaluation.

1. Problem Analysis

In a university, students frequently need information about course registration, class timetables, examination schedules, attendance details, and campus services. Usually, they obtain this information by contacting faculty members, visiting administrative offices, or checking university portals. These methods may take time, and students may find it difficult to get information outside working hours.

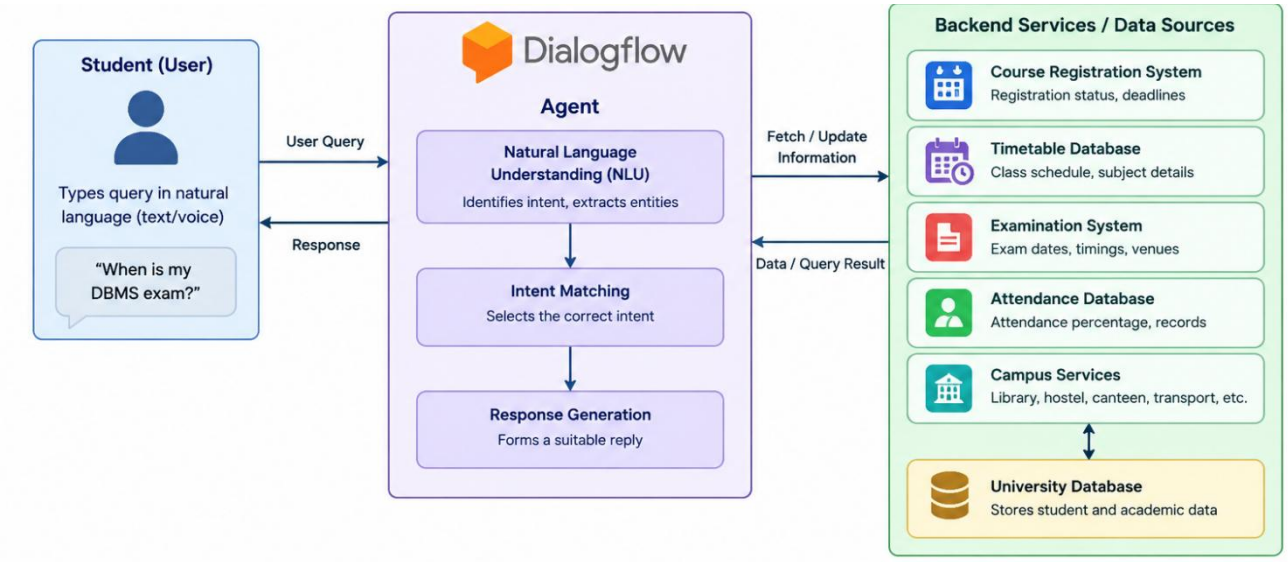
The major problems include:

1. **Delayed Responses:** Students may have to wait to get answers to their queries.
2. **Limited Availability:** Administrative offices may not be available 24/7.
3. **Scattered Information:** Students may need to visit different portals to find academic and campus-related information.
4. **Repetitive Queries:** Faculty members and administrative staff spend time answering the same questions repeatedly.
5. **Difficulty Accessing Information:** Students may struggle to find accurate and up-to-date information quickly.

To overcome these problems, an AI-based Student Support Assistant using Dialogflow is proposed. It allows students to ask questions in natural language and receive appropriate responses. The agent uses Natural Language Understanding (NLU) to identify the user's intention and provide relevant information. For example, when a student asks, "When is my examination?", the agent identifies the examination-related intent and provides the available schedule details.

The proposed system aims to provide quick and consistent responses, reduce the workload of university staff, and improve students' access to academic and campus information. Personal details, such as attendance records, may require database integration and authentication to ensure privacy and accuracy.

Agent Design



1. Student (User)

The student is the main user of the system. They enter a question using natural language through text or voice.

Example: "When is my DBMS exam?"

2. Dialogflow Agent

Dialogflow processes the student's query through three main components:

- Natural Language Understanding (NLU): Understands the meaning of the query, identifies the user's intent, and extracts important details such as the course name.
- Intent Matching: Selects the appropriate intent, such as Exam.Schedule, based on the student's question.
- Response Generation: Prepares a suitable response using the configured answers or information received from backend services.

3. Backend Services and Data Sources

Backend services provide the information needed to answer student queries. They may connect to different university systems, including:

- Course Registration System: Provides registration procedures and deadlines.
- Timetable Database: Provides class schedules and subject details.
- Examination System: Provides exam dates, timings, and venues.
- Attendance Database: Provides attendance percentages and records.
- Campus Services: Provides information about library timings, hostels, canteens, and transportation.

4. University Database

The university database stores relevant student and academic information. Backend services access the required records when necessary. Personal information, such as attendance details, should be provided only after appropriate authentication and authorization.

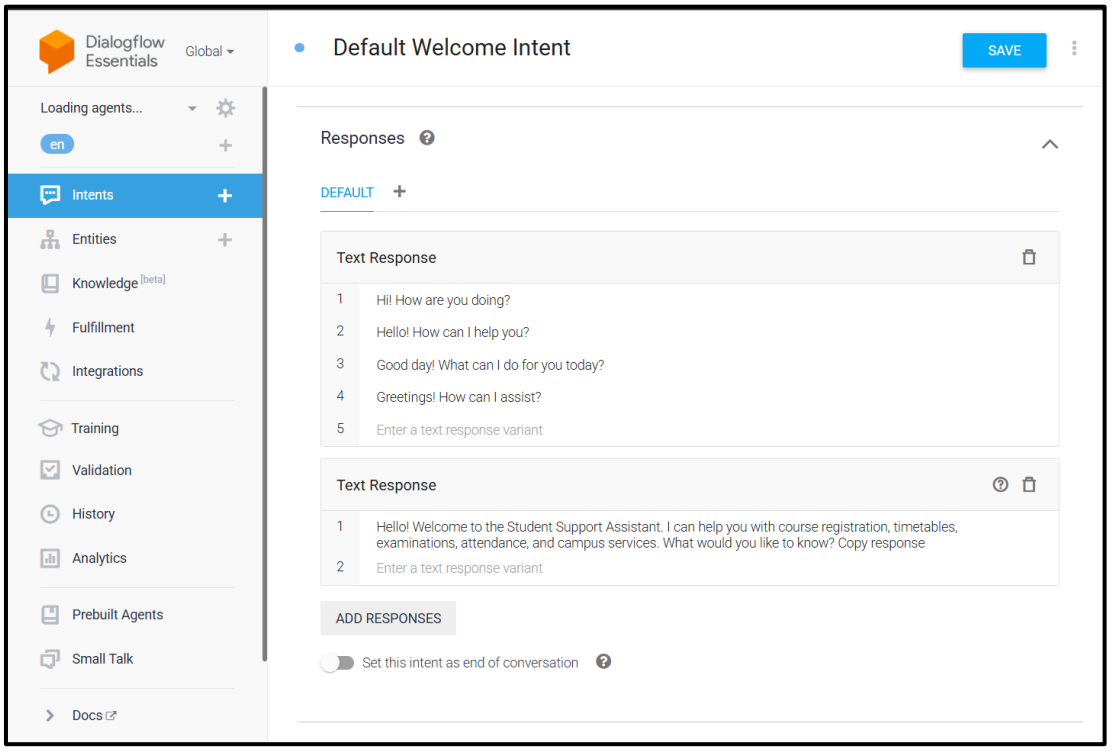
5. Response Delivery

After processing the query and obtaining the necessary information, Dialogflow sends the response back to the student.

Example response: “Your DBMS examination is scheduled for 20 October at 10:00 AM.”

2. Dialogflow Agent Development

2.1 Agent Setup



2.2 Intents and Training Phrases

Intents represent the different purposes behind student queries. Each intent contains training phrases that help Dialogflow recognize what the student wants.

Intent Name	Sample Training Phrases	Response
Default Welcome	Hi, Hello, Good morning	Hello! Welcome to the Student Support Assistant. How can I help you?

Course.Registration	How do I register for courses?; What is the registration deadline?	You can register for courses through the university student portal.
Timetable	Show my timetable; What classes do I have tomorrow?	Please specify the day or semester to get your timetable information.
Exam.Schedule	When is my DBMS exam?; Show the exam timetable.	Please provide the course name to check the examination schedule.
Attendance.Check	Check my attendance; What is my attendance percentage?	Please follow the university's authorized attendance-checking process.
Campus.Services	What are the library timings?; Where is the medical centre?	I can help you with library, hostel, canteen, transport, and other campus services.
Default Fallback	An unrecognized or unrelated query	Sorry, I couldn't understand your question. Please ask about registration, timetables, examinations, attendance, or campus services.
Goodbye	Bye; Thank you	You're welcome! Have a great day.

These are sample responses. Actual deadlines, timetables, examination schedules, and attendance records must be configured with verified university information or connected data services.

2.3 Entities

Entities help Dialogflow identify important information within a student's query.

Entity	**Type**	**Example Values**
@course_name	Custom entity	AI, DBMS, Java, Computer Vision
@department	Custom entity	CSE, ECE, Mechanical, Civil
@service_type	Custom entity	Library, Hostel, Canteen, Transport
@sys.date	System entity	Tomorrow, 20 October
@sys.number	System entity	Semester number or other numeric input

For example, in the query “When is my DBMS exam?”, Dialogflow can identify Exam.Schedule as the intent and DBMS as the @course_name entity.

2.4 Contexts and Follow-up Intents

Contexts help the agent maintain relevant information across multiple messages. They allow students to ask follow-up questions without repeating all the details.

Context	Purpose	Example
exam-followup	Maintains examination-related context	Student: “When is my AI exam?” Student: “Where is it?”
timetable-followup	Maintains timetable-related context	Student: “Show Monday's timetable.” Student: “What about Tuesday?”
awaiting-regno	Continues an attendance query when an authorized identifier is needed	Student: “Check my attendance.” Agent: “Please enter your register number.”

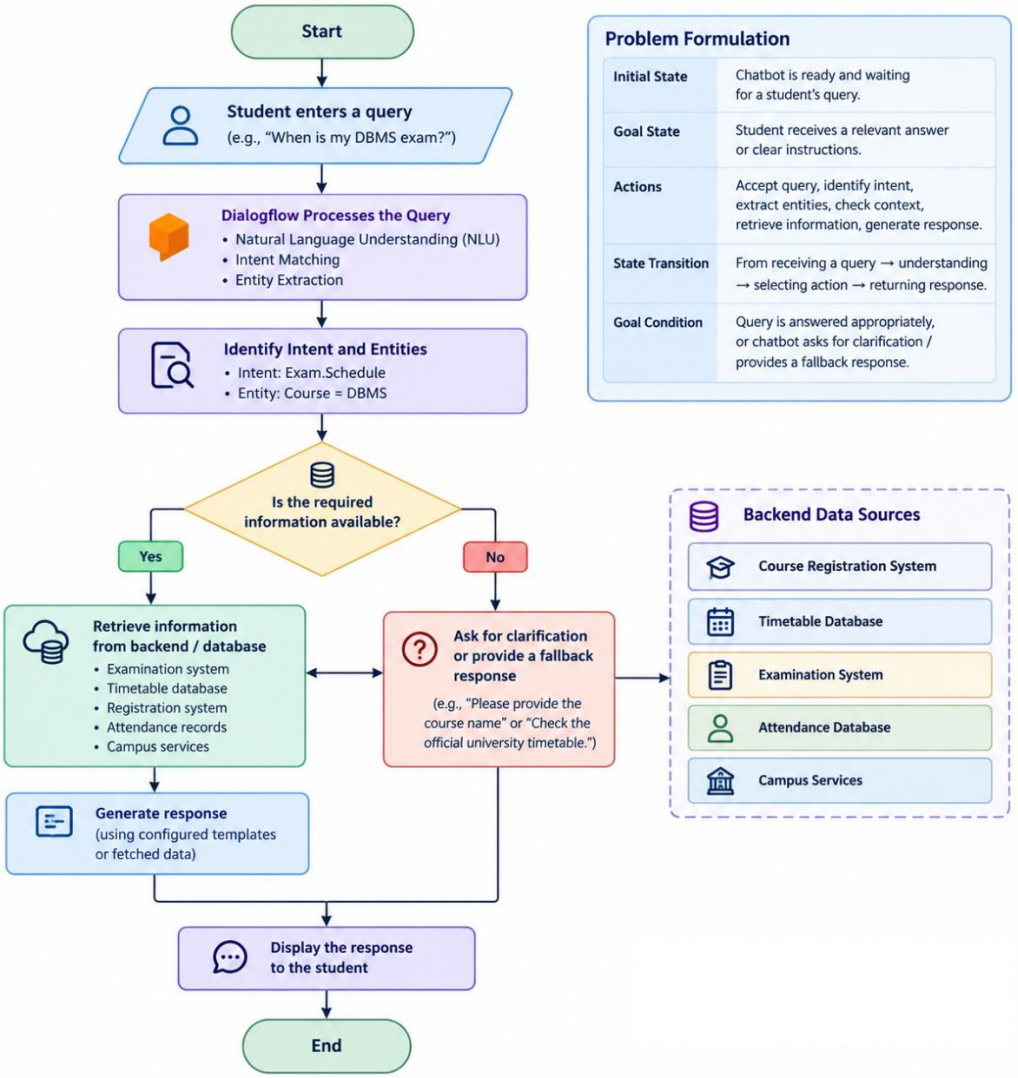
3. Problem Formulation and Decision Flow

3.1 Problem Formulation

The Student Support Assistant is designed to answer common university-related questions. The problem is formulated by defining the initial state, goal state, available actions, and state transitions.

Component	Description
Initial State	The chatbot is ready and waiting for a student's query.
Goal State	The student receives a relevant answer or clear instructions to find the required information.
Actions	Accept query, identify intent, extract entities, check context, retrieve information if available, and generate a response.
State Transition	The system moves from receiving a query to understanding it, selecting an action, and returning a response.
Goal Condition	The query is answered appropriately, or the chatbot asks for clarification or provides a fallback response.

3.2 Flow Diagram



4. Search Strategy Application

4.1 Introduction

A search strategy helps an intelligent system identify a suitable path from the user's initial query to the required answer. In the Student Support Assistant, students may ask questions about course registration, class timetables, examination schedules, attendance, and campus services.

Dialogflow uses natural language understanding and intent matching to identify the purpose of a student's query. A dialogue flow can also be represented as a search tree, where each node represents a conversation state and each branch represents a possible action or response.

4.2 Search Strategy: Breadth-First Search (BFS)

BFS explores a search tree level by level, starting from the initial node and examining its neighboring nodes before moving to deeper levels. It is useful for demonstrating how a student query could navigate through a structured set of conversation states.

Steps involved:

1. Start from the initial state when the student enters a query.
2. Identify the query category, such as exams, attendance, or registration.
3. Explore the relevant conversation path and check whether more information is needed.
4. Select the appropriate response or request clarification.
5. End the interaction when the query is answered or suitable guidance is provided.

4.3 Example

Suppose the student enters: “When is my DBMS exam?”

Step	Search state	Action
1	Start	Receive the student's query.
2	Query analysis	Identify the examination-related request.
3	Intent selection	Match Exam.Schedule.
4	Entity extraction	Identify DBMS as the course.
5	Response selection	Provide available exam information or direct the student to the official timetable.

4.4 Why This Strategy Is Suitable

- Organized navigation: Conversation states can be arranged into a structured flow.
- Clear decisions: The system can distinguish between different query categories.
- Clarification: Missing details can be requested before giving an answer.
- Fallback handling: Unrecognized queries can receive a fallback response.
- Improved interaction: Dialogflow contexts can help match follow-up questions according to the ongoing conversation.

5. Testing and Evaluation

5.1 Introduction

Testing and evaluation are performed to verify whether the Student Support Assistant correctly understands student queries and provides relevant responses. The chatbot is tested with different questions related to course registration, timetables, examination schedules, attendance, and campus services. Unrecognized queries are also tested to check whether the fallback response works correctly.

5.2 Test Cases

The following test cases can be used to evaluate the chatbot in Dialogflow.

Test Case	**Student Query**	**Expected Result**
TC01	Hi	Displays a welcome message.
TC02	How do I register for courses?	Provides course registration guidance.
TC03	Show my timetable.	Requests timetable details or provides available information.

TC04	When is my DBMS exam?	Identifies the examination intent and course name.
TC05	Check my attendance.	Provides authorized attendance-checking instructions.
TC06	What are the library timings?	Provides campus service guidance.
TC07	Tell me something unrelated to university services.	Displays a fallback response.
TC08	Thank you, goodbye.	Provides an appropriate closing response.

5.3 Evaluation Criteria

The chatbot can be evaluated using the following criteria:

- Intent Recognition: Whether the correct intent is identified for the student's query.
- Response Quality: Whether the response is relevant, clear, and understandable.
- Entity Extraction: Whether important details, such as course names, are identified correctly.
- Fallback Handling: Whether the chatbot handles unsupported or unclear queries appropriately.
- Response Consistency: Whether similar queries receive consistent responses.

5.4 Result Analysis

After executing the test cases, compare the actual responses with the expected results. If an incorrect intent is detected, improve the training phrases. If a query is not understood, add more examples or refine the fallback response. If the chatbot provides general guidance instead of actual examination or attendance information, connect it to the appropriate university data source.

You can calculate intent recognition accuracy using:

$$\text{Accuracy} = \frac{\text{Correctly identified intents}}{\text{Total test queries}} \times 100$$

For example, if 18 out of 20 test queries are assigned the correct intent, the intent recognition accuracy is 90%. This is only an example, not an actual result from your chatbot.

5.5 Improvements

The system can be improved by:

1. Adding more training phrases for different ways students ask questions.
2. Improving entity recognition for course names, dates, and departments.
3. Refining fallback responses and follow-up questions.
4. Integrating official university systems for accurate, up-to-date information.
5. Adding authentication before accessing personal attendance or registration records.

5.6 Conclusion

Testing and evaluation help determine whether the Student Support Assistant responds appropriately to student queries. By testing different intents, entities, and fallback situations, the chatbot can be refined to provide clearer and more reliable support. Integration with verified university data sources would further improve its usefulness and accuracy.